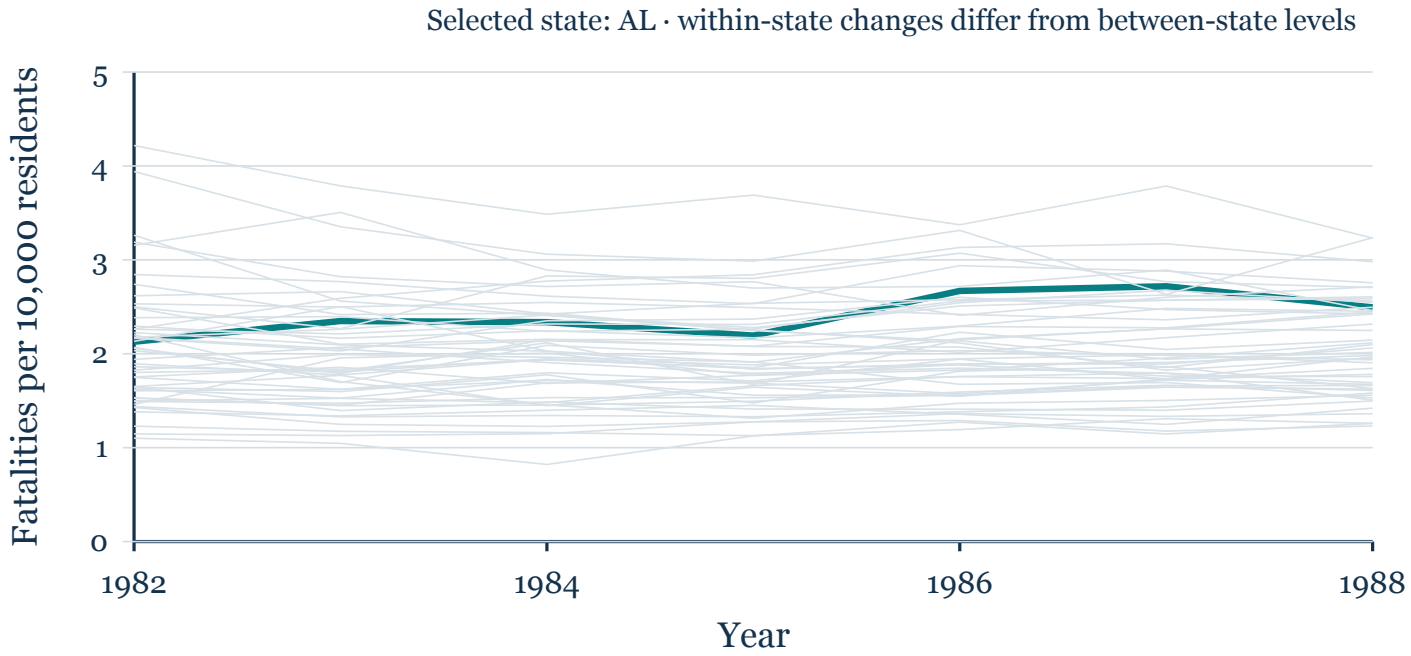
Can repeated observations improve a policy comparison?

Use changes within states to study alcohol taxes and traffic fatalities.



Observed state fatality rates, 1982–1988. The control selects a state; pale paths show the remaining states.

A panel has identities as well as dates

Repeated cross sections do not give the same within-unit comparison.

> One row is a state in a year.

> Before estimating, verify that the state-year pair uniquely identifies rows and inspect the coverage.

Write a stable omitted effect explicitly

An unobserved unit component may be related to policy.

> $Y_{it} = \beta X_{it} + \alpha_i + \lambda_t + u_{it}$

> α_i: stable state component; λ_t: common year shift.

> u_{it}: remaining state-year error.

Follow one unit across two dates

Subtracting the same unit's outcomes removes its stable intercept.

> $(X_{i1}, X_{i2}) = (2, 4), \quad (Y_{i1}, Y_{i2}) = (10, 14)$

> $\Delta X_i = 2, \quad \Delta Y_i = 4, \quad \alpha_i - \alpha_i = 0$

First differencing removes a stable unit component

Subtract the earlier equation from the later equation.

> $$\Delta Y_i = \beta \Delta X_i + \Delta \lambda + \Delta u_i$$

A two-period athlete example uses the same algebra

State what disappears and what the comparison still needs.

> $GPA_{it} = \beta\, season_{it} + \alpha_i + \lambda_t + u_{it}$

> $\Delta GPA_i = \beta \Delta season_i + \Delta\lambda + \Delta u_i$

> GPA: grade point average; season: a zero-or-one indicator.

First differences

Try the comparison before opening the explanation.

Practice

In $GPA_{it} = \beta\, season_{it} + \alpha_i + \lambda_t + u_{it}$ for two semesters, write the differenced equation and say what disappears.

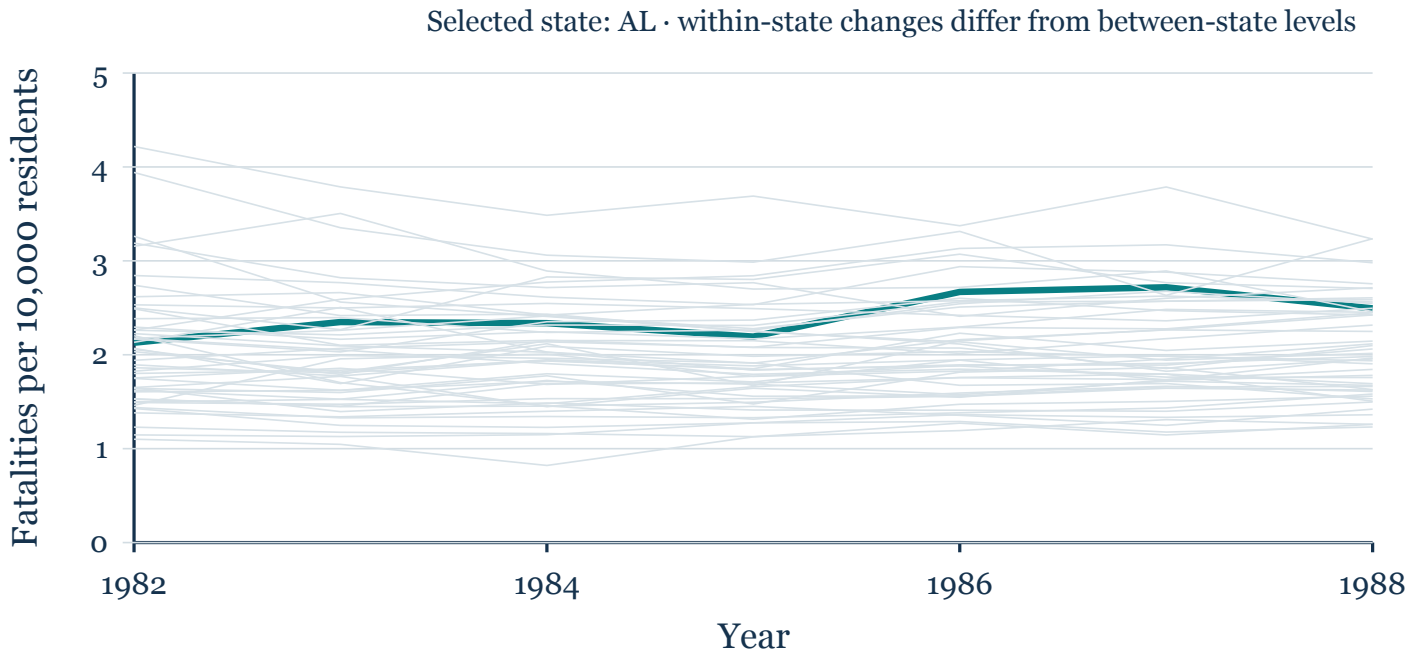
First differences: explanation

Follow the calculation and interpret its units.

> $\Delta GPA_i = \beta \Delta season_i + \Delta\lambda + \Delta u_i$. The time-invariant α_i disappears, as do coefficients on time-invariant regressors. Semester effects do not disappear; with two periods they become a common intercept. Time-varying confounding can remain.

Demeaning uses each unit's own average

Fixed effects remove the same stable component over several periods.



Observed state fatality rates, 1982–1988. The control selects a state; pale paths show the remaining states.

$$X_i = (2, 4) \Rightarrow X_i - \overline{X}_i = (-1, 1)$$

$$Y_i = (10, 14) \Rightarrow Y_i - \overline{Y}_i = (-2, 2)$$

$$Y_{it} - \overline{Y}_i = \beta(X_{it} - \overline{X}_i) + (\lambda_t - \overline{\lambda}) + (u_{it} - \overline{u}_i)$$

Two-period fixed effects and first differences share a slope identity

The identity requires the same two-period sample and specification.

> With two observations per unit, demeaning produces one half of the change and its negative.

> Our application compares a two-period change from 1982 to 1988 with a fixed-effects fit using all seven years.

Read pooled and within-state associations

Removing stable differences can change the sign.

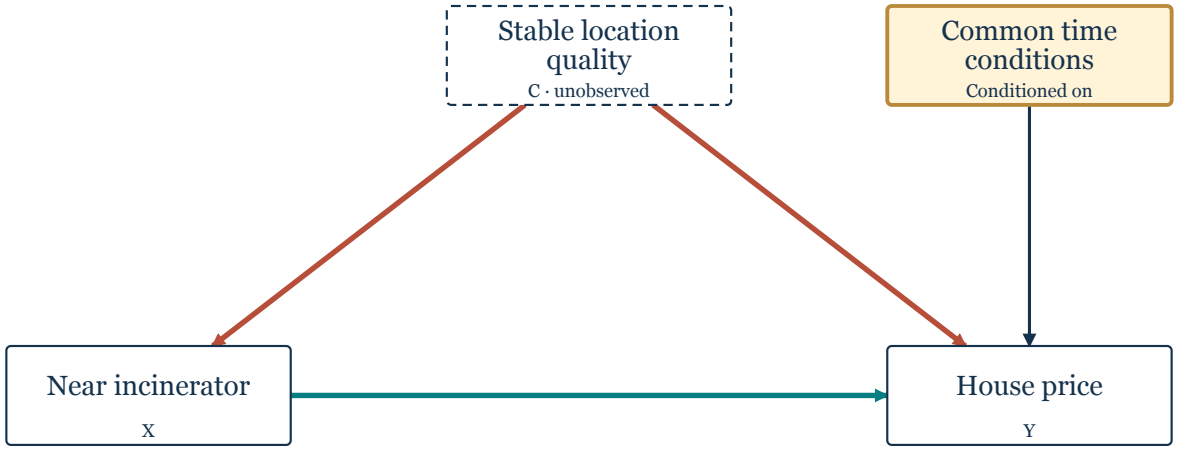
```
. xtreg fatality beertax i.year, fe vce(cluster state)

------------------------------------------------------------------------------
             |               Robust
    fatality | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     beertax |  -.6399799   .3570783    -1.79   0.080    -1.358329     .078369
             |
       _cons |    2.42847   .2016884    12.04   0.000     2.022725    2.834215
-------------+----------------------------------------------------------------
```

Calendar-indicator coefficient rows are omitted here. The output download contains the full numerical results.

Use year effects for common shocks

They do not remove state-specific changes.



1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

> Year effects: common date shifts. State effects: stable additive state differences.

> A location common cause is not removed by a common calendar control.

State the fixed-effects exogeneity condition

Transforming variables does not make their remaining changes random.

> $E(u_{it} \mid X_{i1}, \ldots, X_{iT}, \alpha_i) = 0$ as a sufficient panel condition

Build a difference-in-differences comparison

Begin with the Head Start policy question from the original DiD lesson.

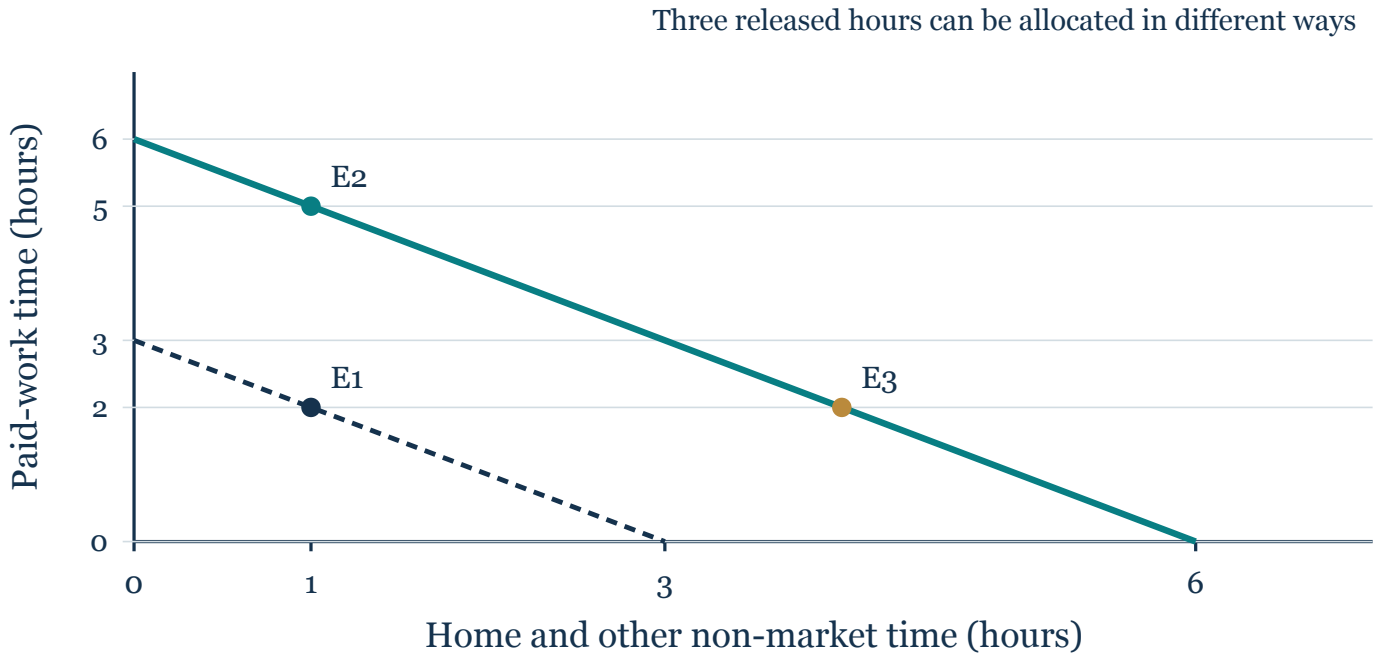
> Policy question: preschool access and maternal employment.

> Define eligibility before the expansion, the comparison population and the two dates.

> Reading: Wikle and Wilson (2023), Access to Head Start and Maternal Labor Supply.

Preschool access changes a household's feasible choices

Released care time can support paid work or another activity.

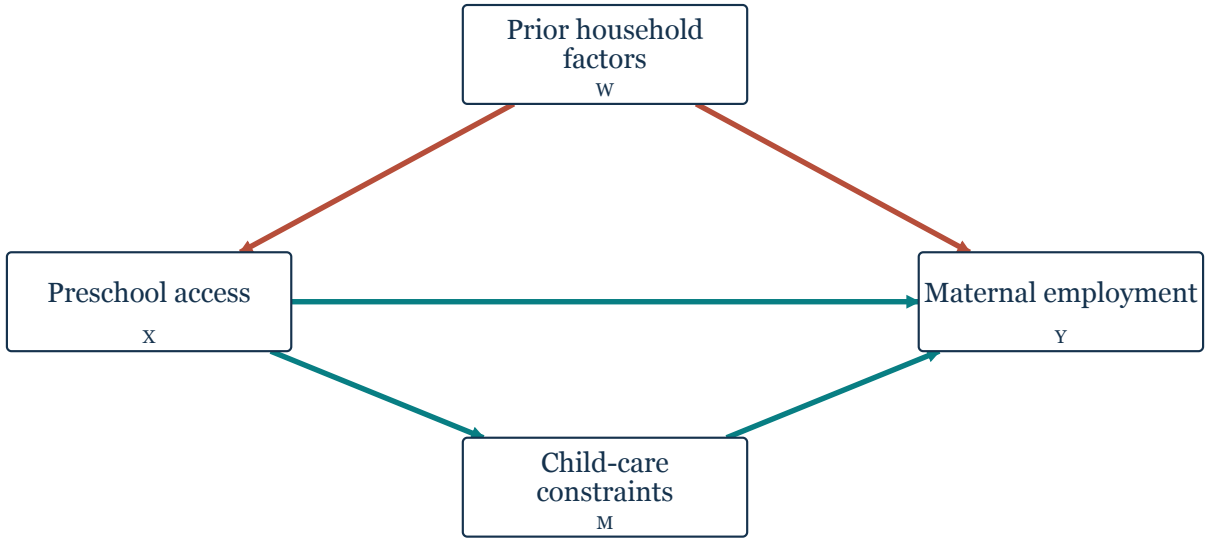


Constructed time-allocation diagram adapted from Qiu's Head Start lesson. Points illustrate feasible choices, not estimated responses.

- Axes: home/other time and paid-work time.
- Preschool can relax a care constraint without forcing a particular response.
- The diagram is a constructed time-allocation illustration.

Eligibility, access and participation are different variables

State the policy exposure whose effect is being studied.



2 open directed routes · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

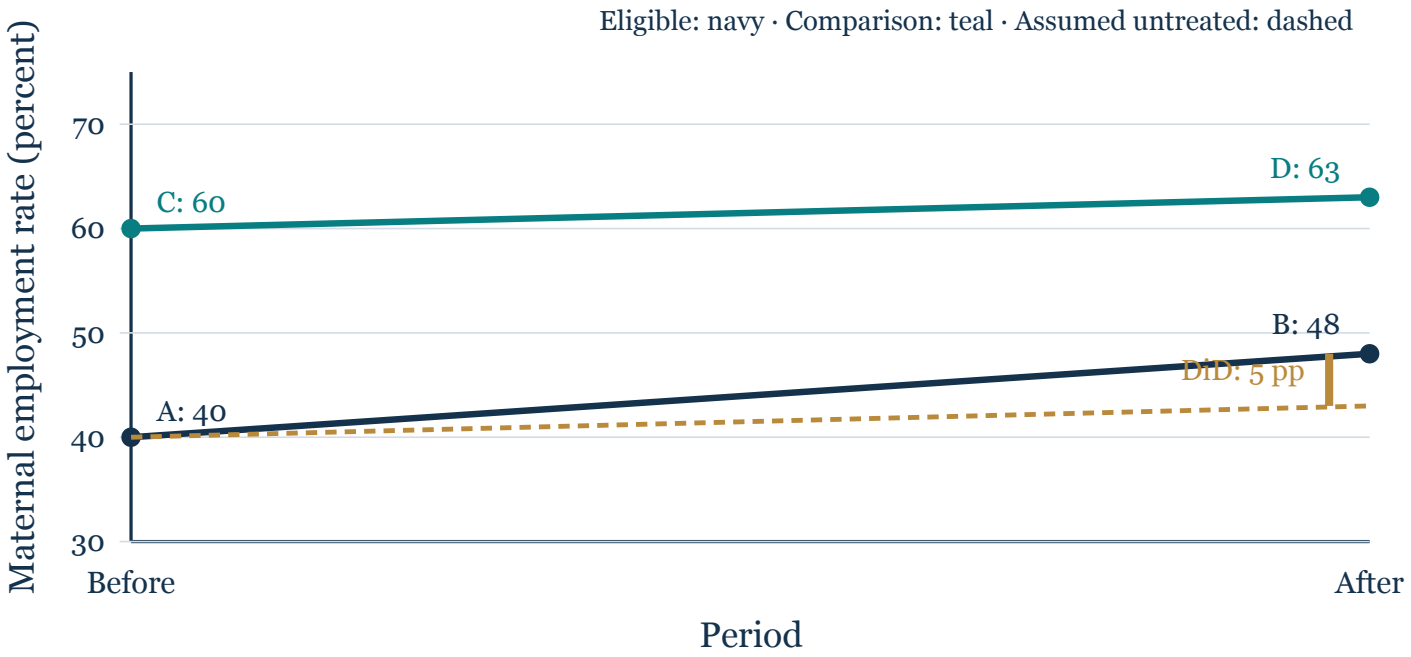
> Eligibility: policy conditions; access: an available place; participation: actual use.

> Define the expansion comparison before interpreting a coefficient as an attendance effect.

> A causal diagram does not establish parallel untreated trends.

Parallel trends is about unobserved untreated changes

It is not a requirement that the groups begin at the same level.



Constructed Head Start employment-rate example using the original A/B/C/D sequence. Dashed untreated paths are assumptions; these are not estimates from the reading.

- Observed DiD: $(Y_B - Y_A) - (Y_D - Y_C) = (48 - 40) - (63 - 60) = 5$ percentage points.
- Assumed untreated eligible post rate: $40 + (63 - 60) = 43\%$.
- Parallel trends concerns the missing no-expansion change, not equal starting levels.

No anticipation and stable comparison units also matter

Define when treatment can begin to affect behaviour.

> No anticipation: the policy has not already changed the baseline outcome.

> Check spillovers and the population represented at both dates.

> A clear policy definition is part of the causal model.

DiD arithmetic

Try the comparison before opening the explanation.

> Work in percentage points before converting to a zero-or-one outcome scale.

Practice

In the Head Start teaching diagram, eligible mothers' employment rises from 40% to 48%; comparison mothers' employment rises from 60% to 63%. Calculate DiD in percentage points and on the zero-or-one employment-regression scale. State the untreated-trend assumption.

DiD arithmetic: explanation

Follow the calculation and interpret its units.

> $(48 - 40) - (63 - 60) = 5$ percentage points.

> Binary-outcome regression contrast: 0.05.

> The untreated eligible post mean is assumed, not observed.

The interaction in a two-by-two DiD equation has a specific meaning

Its interpretation follows the four group-time means.

> $Y_{it} = \beta_0 + \beta_1 P_t + \beta_2 G_i + \beta_3 G_i P_t + u_{it}.$

> G_i: predefined eligible group; P_t: post period; Y_{it}: employment indicator.

> Constructed interaction: $\beta_3 = 0.05$, or five percentage points.

Different characteristics can imply different untreated changes

Unequal starting levels and unequal untreated trends are different issues.

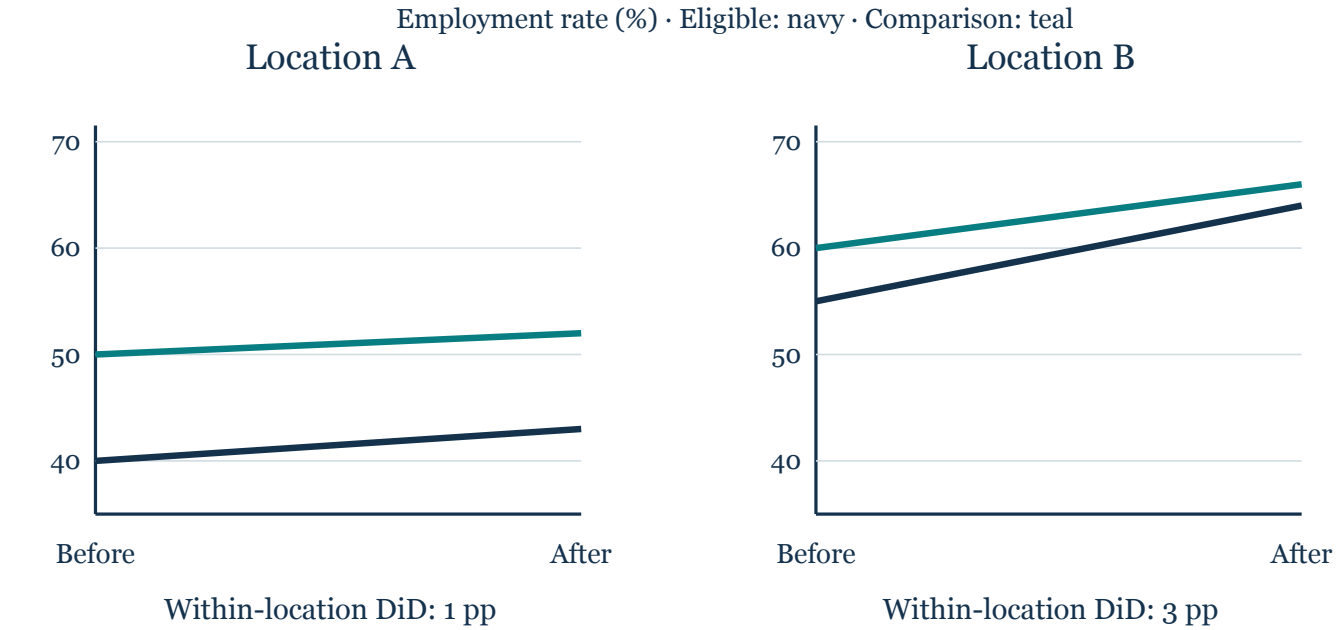
> W: characteristics fixed before intervention.

> Conditional trends: $E[\Delta Y(0) \mid G = 1, W] = E[\Delta Y(0) \mid G = 0, W]$.

> Here the change in Y(0) means the change without expansion.

Compare within location before averaging across locations

The original two-location idea separates changing opportunities from policy effects.

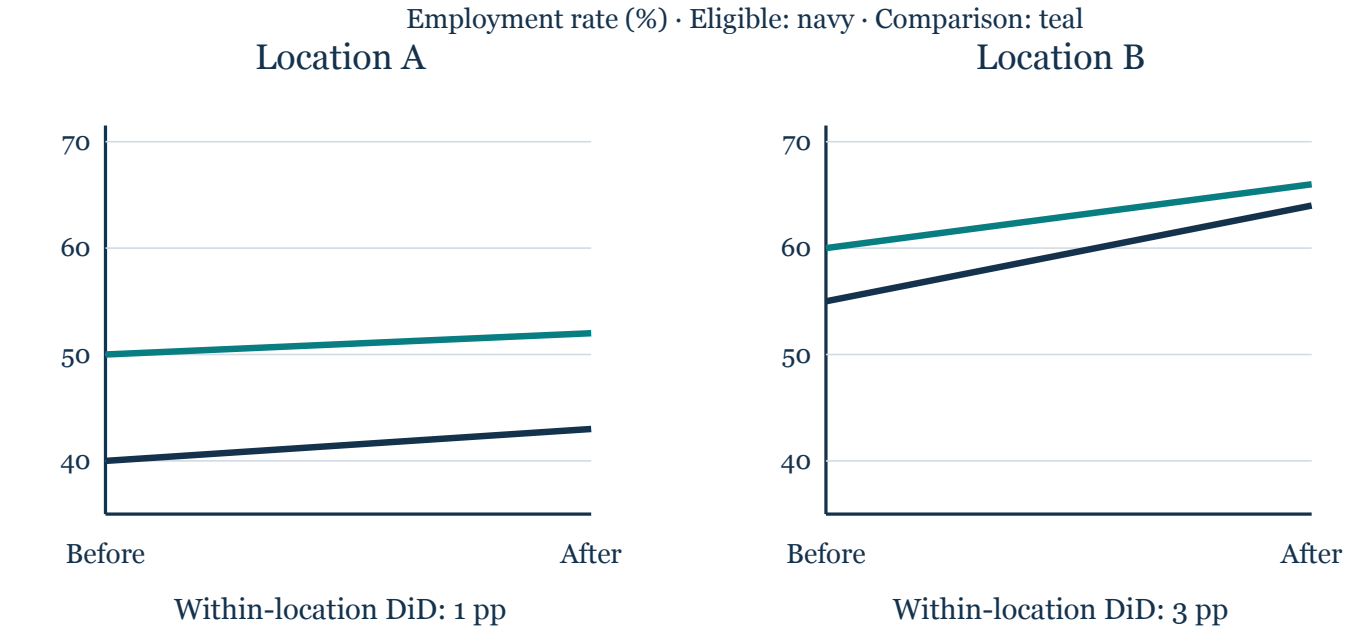


Constructed two-location comparison adapted from the Overleaf lesson. Within-location trends and effects are specified for the illustration.

- Location A: group changes 3 and 2; within-location DiD 1 percentage point.
- Location B: group changes 9 and 6; within-location DiD 3 percentage points.
- Different location mixtures can reverse the unadjusted aggregate comparison.

Average conditional effects for the treated population

The target determines the averaging weights.



Constructed two-location comparison adapted from the Overleaf lesson. Within-location trends and effects are specified for the illustration.

- $\tau(W)$: the conditional effect among treated units at characteristic W.
- $ATT = E[\tau(W) \mid G = 1]$.
- Example: $0.75(1) + 0.25(3) = 1.5$ percentage points.

A covariate in levels need not adjust a trend

Decide whether the model represents untreated changes and effect heterogeneity.

> Time-invariant levels: $\Delta(\gamma W_i) = 0$.

> Model the untreated change as a function of pre-treatment W.

> Allow appropriate heterogeneity and average for the stated population.

Repeat the original zero-effect income-targeting experiment

Known simulation truth lets us diagnose a misleading comparison.

```
. regress dy group, vce(robust)

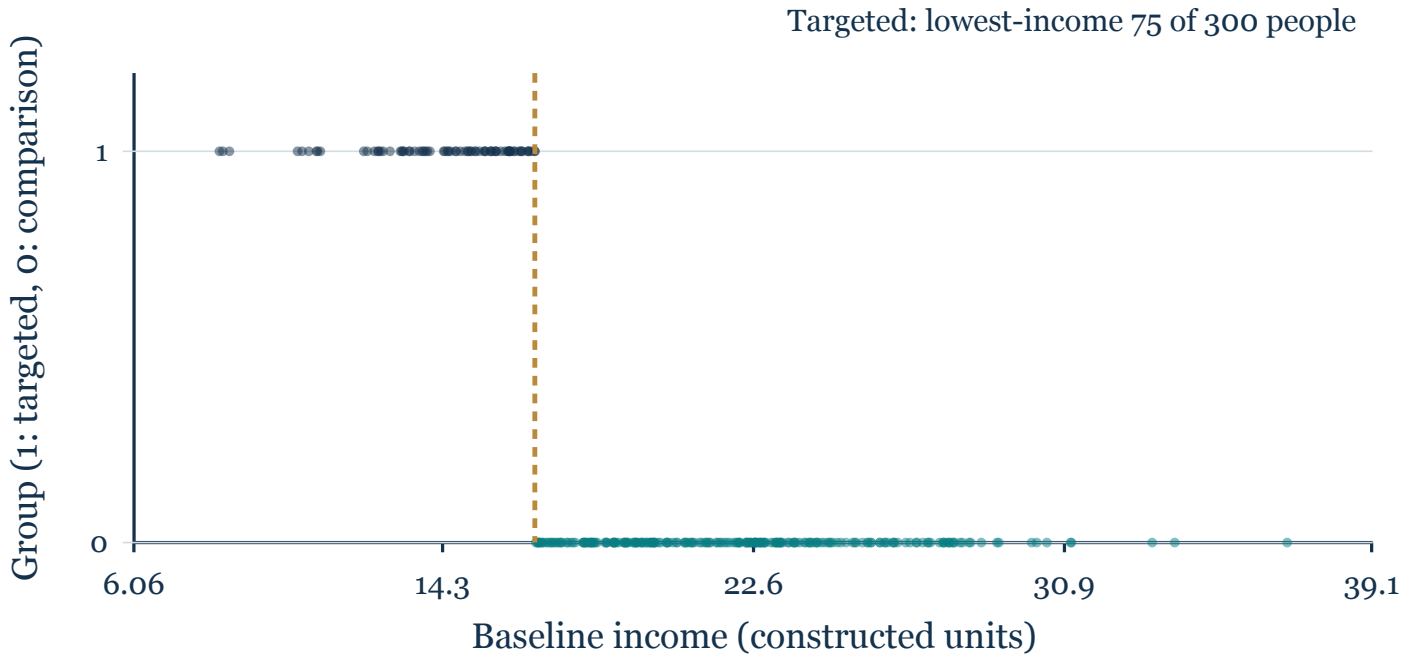
------------------------------------------------------------------------------
             |               Robust
          dy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       group |  -3.222965   .1420669   -22.69   0.000    -3.502547   -2.943384
       _cons |   2.900887   .1026169    28.27   0.000     2.698941    3.102832
------------------------------------------------------------------------------
```

Actual StataMP output. Reproduce with did_overleaf.do; the full output is in the download.

- Constructed data: 300 people; 75 targeted; known effect zero.
- Untreated changes vary with the baseline income used for targeting.
- A correct linear change model and a defensible comparison are different requirements.

Similar-group comparisons need overlap

An estimator cannot create comparison observations absent from the data.



Actual generated observations from the zero-effect 300-person experiment. Lowest-quarter baseline-income targeting leaves no comparison observations below the cutoff.

- Overlap for treated characteristics requires comparison observations: $P(G = 0 \mid W) > 0$.
- Deterministic income targeting leaves no comparison observations below its cutoff.
- A fitted extrapolation should be justified and labelled as such.

Doubly robust DiD combines two supporting models

Two chances for model correctness retain the design assumptions.

> Outcome model: $m_0(W) = E[\Delta Y \mid G = 0, W]$.

> Propensity score: $p(W) = P(G = 1 \mid W)$.

> A logistic membership curve stays between zero and one.

> Double robustness concerns supporting-model correctness; conditional trends and overlap remain assumptions.

Read the real-data example from the Overleaf lesson

Identify the sample and the reported target before interpreting ATET.

```
. drdid re age educ black married nodegree hisp re74, ///

------------------------------------------------------------------------------
             | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATET         |
experimental |
   (1 vs 0)  |  -901.2703   393.6127    -2.29   0.022    -1672.737   -129.8037
------------------------------------------------------------------------------
```

Actual StataMP output. Reproduce with did_overleaf.do; the full output is in the download.

- Retained sample: NSW experimental controls and CPS comparison observations.
- Data-source group membership is not actual training receipt.
- ATET and ATT name the same treated-group average target.

Separate spillovers, treatment versions and anticipation

Use the original SUTVA discussion to define the policy comparison precisely.

> No interference: other units' exposure does not change this unit's potential outcome.

> Well-defined treatment: specify the preschool access or service bundle being compared.

> No anticipation: the baseline outcome is not already affected by the future intervention.

Cluster inference for repeated units

Errors within a state can remain related after fixed effects.

> State fixed effects remove a stable mean component; they do not force the remaining errors in different years to be independent.

> The policy assignment and shared shocks help determine the appropriate clustering unit.

Cluster choice

Try the comparison before opening the explanation.

Practice

A policy is assigned to states and repeated outcomes are observed for many districts over years. Why is clustering by district alone potentially insufficient?

Cluster choice: explanation

Follow the calculation and interpret its units.

> Errors and policy exposure may be shared across districts within a state. Inference should reflect the assignment and dependence structure; state clustering is a starting point when states are independent sampling/assignment clusters. Few states require special care, and a universal cluster-count threshold is not a theorem.

Interpret a feasible tax difference

Use the fitted contrast and preserve the outcome units.

```
. lincom .10*beertax

------------------------------------------------------------------------------
    fatality | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   -.063998   .0357078    -1.79   0.080    -.1358329    .0078369
------------------------------------------------------------------------------
```

Inspect the 1982-to-1988 changes separately

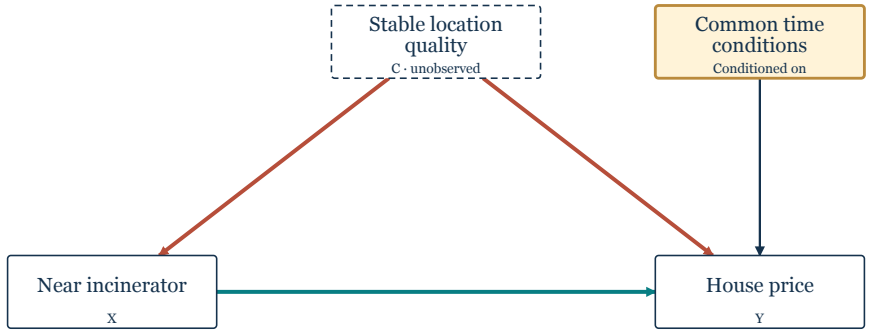
A longer change can use different variation from annual fixed effects.

```
. regress dy dx, vce(robust)

------------------------------------------------------------------------------
             |               Robust
          dy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
          dx |  -1.040973   .3550061    -2.93   0.005    -1.755563   -.3263823
       _cons |  -.0720371   .0653552    -1.10   0.276    -.2035903    .0595161
------------------------------------------------------------------------------
```

DAG practice: pooling is not within-location identification

Trace the routes before choosing a control.



1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Practice

In the incinerator example, common time conditions affect prices while stable location quality affects proximity and prices. Do year indicators close the location route? If proximity never changes, what happens to its level coefficient under unit fixed effects, and what extra comparison could support a DiD question?

DAG practice: pooling is not within-location identification: explanation

Explain what conditioning changes in this stated graph.

> Year indicators do not block $X \leftarrow C \rightarrow Y$ through stable location quality. Unit fixed effects can cancel an additive stable location component, but they also absorb a proximity indicator that never changes within a unit. Its separate level coefficient is then unavailable. A proximity-by-post-period interaction can vary across groups and dates. Interpreting that interaction as a DiD effect requires a well-defined intervention and parallel untreated trends, along with the other stated comparison assumptions. Neither pooling nor the causal diagram alone establishes those assumptions.

Distinguish within-state and endpoint comparisons

Reproduce the evidence, then explain its economic meaning.

Practice

Change the endpoint first-difference comparison to 1983 and 1988. Use the same 48 states, calculate the differences yourself and compare the slope with the original endpoint window. Keep your commands, log, output and interpretation.

Distinguish within-state and endpoint comparisons: explanation

A reproducible result needs both a calculation and an interpretation.

> State FE slope: −0.639980; 1982–1988 endpoint slope: −1.040973.

> 1983–1988 endpoint slope: −0.845389, robust SE 0.332000.

> A changed endpoint window changes the dated comparison.

Next, seek a different source of treatment variation

Panel adjustment may leave time-varying endogeneity.

> Panel methods remove stable unit effects and let us form explicit change comparisons.

> The next lesson introduces instrumental variables, using an outside source of variation that moves the regressor under specified conditions.